

Finley Desai

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Education

Stanford University

Palo Alto, CA

B.S. / M.S. Electrical Engineering, Coterminous Program – continuously enrolled through Jun 2028

M.S. Electrical Engineering – Expected Jun 2028 (graduate coursework in progress, 2026–27)

B.S. Electrical Engineering, Minor in Computer Science – Expected Jun 2027 | GPA: 3.99/4.0

Selected Coursework: Digital System Design, Circuits I-II, Signals and Systems I-II, Computer Systems, Feedback Control Design

2026-27 Coursework: Analog Integrated Circuit Design, Advanced IC Design, VLSI Systems, VLSI Design Projects I-II, Analog Communications Design Lab, Engineering Electromagnetics, Digital Systems Architecture, Computer Systems Architecture

Technical Experience and Projects

Stanford Tambe Lab

Palo Alto, CA

Research Intern (Part-Time), Robotics Chip-on-Board Module

Jun 2026 – Present

- Created Altium library footprints and bond-pad land patterns for the custom die, drawing die-attach and wire-bond geometry directly in layout.
- Designing a ~16-layer chip-on-board module in Altium for on-robot inference: board-level schematic through layout, integrating ~50 ICs including wire-bonded bare die.
- Developing the controlled-impedance stackup, microvia/blind-via architecture, and DDR3-speed routing constraints, using pre-layout simulation in MATLAB Signal Integrity Toolbox to close impedance and timing budgets before fabrication.

Stanford Accelerate Lab

Palo Alto, CA

Research Intern, Hardware Accelerator Test PCB ([GitHub](#))

May 2026 – Jun 2026

- Designed a board-level bring-up and test PCB in KiCad for a custom accelerator ASIC, integrating JTAG and peripheral interfaces and closing clock routing for operation up to 600 MHz.

Stanford Magnetic Resonance Systems Research Lab

Palo Alto, CA

Research Intern

Jun 2025 – Jan 2026

Custom High-Res Wireless Quadrature Digitizer PCB for MRI Radar Prototyping ([Conference Paper](#) | [GitHub](#))

- Designed a dual-channel quadrature digitizer with high-resolution ADCs and a custom analog front-end for continuous-wave radar vital signs monitoring in MRI; measured an ~80 dB improvement in dynamic range over SDR-based acquisition.
- Took the board from schematic and multilayer layout through hand assembly and bring-up, validating the analog signal chain with oscilloscopes, spectrum analyzers, signal generators, and multimeters.
- Developed MicroPython/PIO firmware on a Raspberry Pi Pico W for synchronized acquisition and Bluetooth streaming.

Stanford Molecular Imaging Instrumentation Laboratory

Palo Alto, CA

Research Intern

Mar 2024 – Jun 2025

- Developed and ran custom Monte Carlo physics simulations, analyzing large datasets to design and validate inter-crystal scatter correction and event-positioning algorithms for PET imaging.
- Two conference papers accepted for presentation at IEEE Nuclear Science Symposium & Medical Imaging Conference (NSS/MIC), first author on one; abstracts published in conference proceedings: [Event Positioning](#), [Inter-Crystal Scattering](#).

FPGA Digital Systems Projects – Verilog

- Designed and implemented FPGA digital systems in Verilog, including a bike-light controller and a polyphonic music player with harmonics and envelope modulation; validated each with simulation testbenches.

Stanford Dept. of Electrical Engineering – Undergrad Section Leader, EE 101A (Circuits I)

Sep 2026 – Dec 2026

- Lead weekly lab of ~15 students: bench instrumentation, live debugging of student circuits, lab report grading, office hours.

Additional Experience

American Industrial Partners – Operations Intern

Jun 2026 – Sep 2026

- Built an inventory-aging model for pulp woodyards estimating pine-chemical losses; presented to mill managers.

Technical Skills

Languages/Tools: Verilog, C, C++, Python, MATLAB, Altium Designer, KiCad, LTspice, Git, CMake, Linux

Hardware & Lab: Board-level design/layout/assembly, bring-up & debug, scopes, logic/spectrum analyzers, multimeters, SI analysis